LECTURE 9.1

Laminar Flow Prediction

From Burgers to incompressible Navier–Stokes: the first vector-valued, coupled, genuinely nonlinear problem in the course

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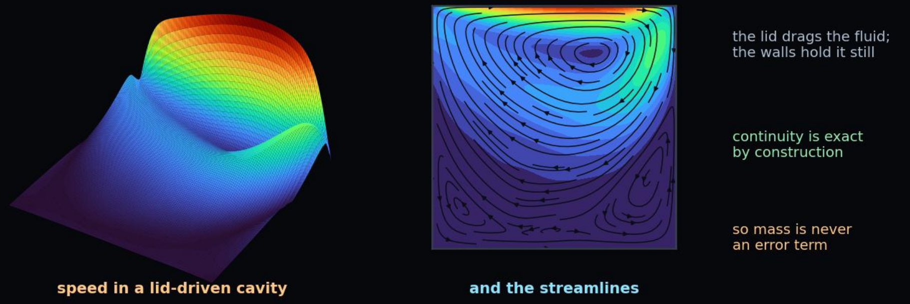
Raissi, Perdikaris & Karniadakis (2019) · Liu (2025) for the machinery



What You Will Be Able To Do

THE GOALS, IN PLAIN WORDS

- **write** — the three residuals, and count the derivatives each one needs
- **explain** — what the Reynolds number sets, both physically and for the method
- **use** — a stream function, and say what it buys and what it costs
- **handle** — a pressure field that is only determined up to a constant
- **place** — collocation points where flow is actually difficult
- **state** — the case for a PINN over a finite-volume code, narrowly

WHAT TODAY ASSUMES

from L7.2	Burgers, and the convection nonlinearity
from L8	scaling, hard enforcement, verification
from L8.2	time as a coordinate

FOUR THINGS CHANGE AT ONCE

Three coupled fields, a nonlinear term, a constraint with no time derivative, and a pressure that is not physically determined.

AND ONE RESULT WORTH THE HOUR

Pressure recovered from velocity alone, with the viscosity, from noisy data.

ORIENTATION

Four Things Change at Once

	L8 — HEAT	L9 — FLOW
Unknown	*one scalar T*	*three fields: u, v, p*
Equations	*one PDE*	*two momentum + one constraint*
Nonlinearity	*none (or mild)*	*convection $(u\cdot\nabla)u$*
Coupling	—	*pressure couples both momenta*
Constraint	—	*incompressibility, no time derivative*
Difficulty knob	*Bi, Fo*	*Reynolds number*

The machinery is still the machinery. What is new is that the network now has three outputs, the residual has three components, and one of them is a constraint rather than an evolution equation.

Recap L8; Raissi et al. (2019)

FORMULATION

Incompressible Navier–Stokes

MASS CONSERVATION — A CONSTRAINT, NOT AN EVOLUTION EQUATION

$$\nabla \cdot \mathbf{u} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

MOMENTUM CONSERVATION

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\frac{1}{\rho}\nabla p + \nu\, \nabla^2 \mathbf{u}$$

- four terms in the momentum equation, each with a physical meaning
- the convection term is the source of every difficulty in this lecture
- it is nonlinear, and it couples the components
- **continuity has no time derivative and no source** — it constrains rather than evolves
- delete convection and you have Stokes flow, which behaves like the heat problems

Standard fluid mechanics; Raissi et al. (2019)

The Residual, Component by Component

X-MOMENTUM

$$u_{,t} + u\,u_{,x} + v\,u_{,y} = -\frac{1}{\rho}p_{,x} + \nu\left(u_{,xx} + u_{,yy}\right)$$

Y-MOMENTUM

$$v_{,t} + u\,v_{,x} + v\,v_{,y} = -\frac{1}{\rho}p_{,y} + \nu\left(v_{,xx} + v_{,yy}\right)$$

CONTINUITY

$$\nabla \cdot \mathbf{u} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

- three residuals, all from the same graph, all at the same points
- **count the derivatives** — one in time, two first in space, two second
- the products are formed on the graph, and autograd handles the nonlinearity exactly

Raissi et al. (2019), NavierStokes.py

The Reynolds Number Is the Difficulty Knob

DEFINITION

$$Re = \frac{UL}{\nu} = \frac{inertia}{viscosity}$$

NON-DIMENSIONAL FORM — RE IS THE ONLY PARAMETER

$$\mathbf{u}^*_{,t} + (\mathbf{u}^* \cdot \nabla)\mathbf{u}^* = -\nabla p^* + \frac{1}{Re}\nabla^2 \mathbf{u}^*$$

- non-dimensionalise and every material property collapses into one number
- two flows with the same Reynolds number behave identically
- and it measures how hard the problem is for a PINN
- low Re is smooth; high Re is thin layers and sharp gradients
- the practical ceiling on the raw equations is a few hundred

REGIMES PAST A CYLINDER

$Re < 5$	creeping, attached
$5 – 40$	steady twin vortices
$40 – 200$	vortex shedding
$200 – 10^5$	wake turbulent
$> 10^5$	fully turbulent

A plain PINN is comfortable in the first three bands and increasingly unreliable beyond them.

2-D Burgers: Navier–Stokes Without the Pressure

X-COMPONENT

$$u_{,t} + u\,u_{,x} + v\,u_{,y} = \nu\left(u_{,xx} + u_{,yy}\right)$$

Y-COMPONENT

$$v_{,t} + u\,v_{,x} + v\,v_{,y} = \nu\left(v_{,xx} + v_{,yy}\right)$$

- take Navier–Stokes, delete the pressure gradient, drop incompressibility
- what is left is 2-D Burgers
- it keeps the convection nonlinearity and the vector coupling
- **so it is the stepping stone** — and Ex_9.1 starts here
- if your Burgers solver works, adding pressure is incremental

WHAT EACH MODEL KEEPS

	BURGERS	N–S
convection (u·∇)u	yes	yes
viscous diffusion	yes	yes
vector coupling	yes	yes
pressure gradient	no	yes
incompressibility	no	yes

the two new difficulties arrive first, alone

L7.2 slides 23–27; Raissi et al. (2019)

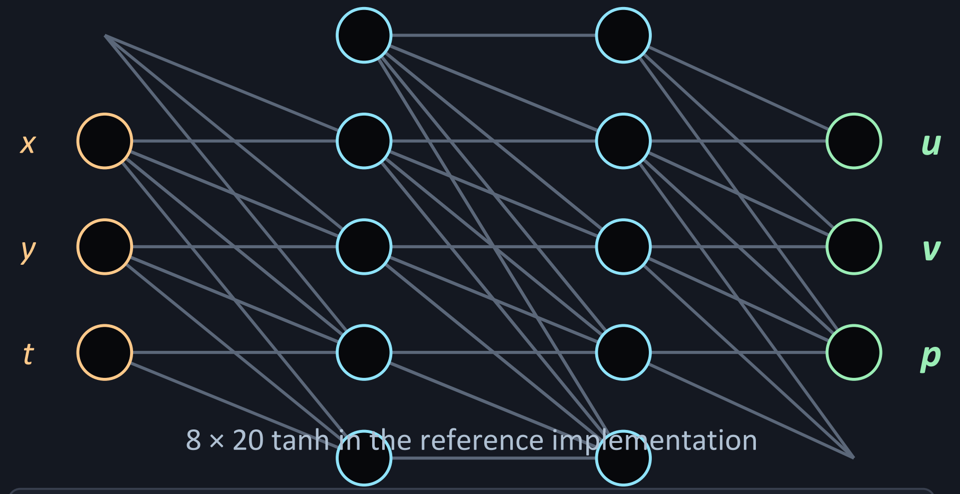
One Network, Three Outputs

THE ANSATZ

$$(u,\ v,\ p) = \mathcal{N}_{\widehat{w}}(x, y, t)$$

- one network with three outputs, not three networks
- the fields share hidden representations, which is the point
- pressure enters only through its gradient, so its absolute level is never needed
- depth matters more here than in the heat problems
- **tanh throughout** — the residual needs second derivatives

SHARED TRUNK, THREE HEADS



Three outputs, three residuals — but only two of them are evolution equations. The third constrains the first two.

Three Ways to Impose Incompressibility

PENALTY

Add $\|\nabla\cdot u\|^2$ as another loss term.

Simplest to code.
Only approximately satisfied, and it competes with the momentum residuals for the optimiser's attention.

STREAM FUNCTION

Predict ψ, take u and v from its derivatives.

Divergence-free by construction, to machine precision.
One fewer output and one fewer loss term.
2-D only. Raissi's choice.

PROJECTION

Correct the velocity onto a divergence-free space.

Standard in classical CFD.
Awkward inside a PINN — it needs a Poisson solve, which is exactly what we were trying to avoid.

STREAM FUNCTION DEFINITION

$$u = \frac{\partial \psi}{\partial y}, \qquad v = -\frac{\partial \psi}{\partial x}$$

WHY IT WORKS: MIXED PARTIALS COMMUTE

$$\nabla \cdot \mathbf{u} = \psi_{,yx} - \psi_{,xy} \equiv 0$$

Raissi et al. (2019); standard practice

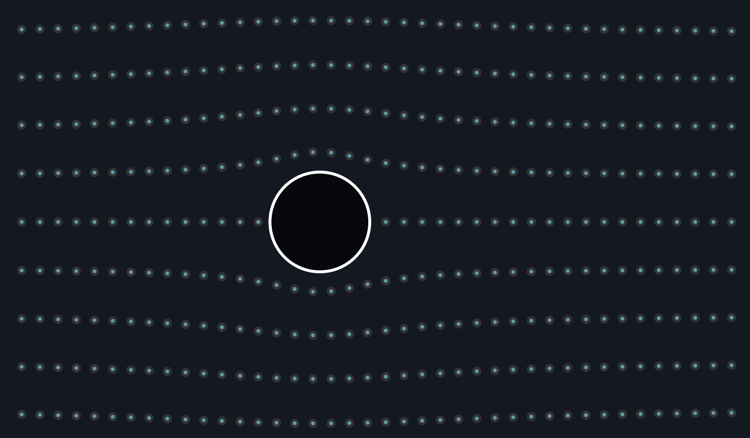
The Stream Function in Practice

VELOCITY FROM Ψ

$$u = \frac{\partial \psi}{\partial y}, \qquad v = -\frac{\partial \psi}{\partial x}$$

- the network predicts the stream function and the pressure
- velocities come from differentiating it
- continuity is then satisfied structurally, and its loss term disappears
- the cost is third derivatives in the momentum residual
- **and the limitation is real** — a stream function exists only in 2-D or axisymmetric flow

CONTOURS OF Ψ ARE STREAMLINES



$\nabla \cdot u = 0$ holds exactly, at every point, always

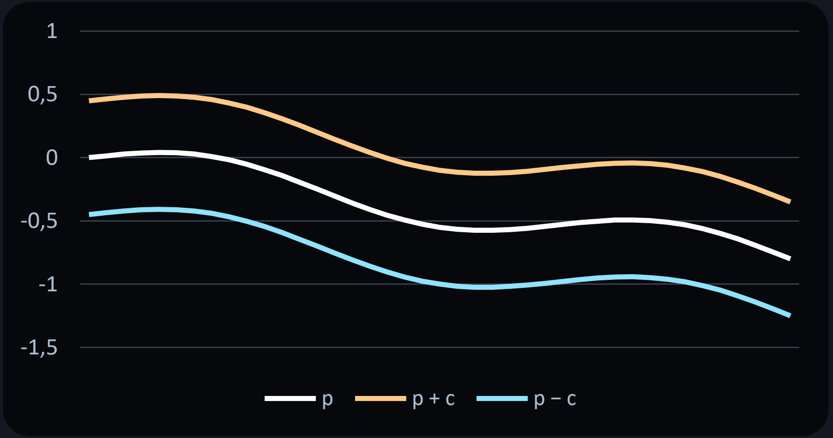
Pressure Is Only Determined Up to a Constant

THE GAUGE FREEDOM

$$p \to p + c \quad \Rightarrow \quad \text{same velocity field}$$

- in incompressible flow the pressure has no thermodynamic meaning
- it appears only through its gradient, so any constant may be added
- the same degeneracy as the pure-Neumann heat problem in L8.1
- so never compare absolute pressures between runs
- **fix it by prescribing zero at one point** — it costs nothing

SAME FLOW, THREE PRESSURE FIELDS



identical gradients — identical velocity

Incompressible flow theory; cf. L8.1 slide 7

FORMULATION

Boundary Conditions for a Flow Problem

NO-SLIP WALL

$$\mathbf{u} = \mathbf{0} \quad \text{on the wall}$$

Both velocity components vanish on a solid surface. The single most important condition in viscous flow — it is what creates boundary layers, drag and wakes.

INLET

$$u = U(y), \quad v = 0 \quad \text{at the inlet}$$

A prescribed velocity profile. Uniform is simplest; fully developed Poiseuille is more realistic for a pipe and avoids a spurious entrance region.

OUTLET

$$p = 0, \qquad \frac{\partial u}{\partial n} = 0 \quad \text{at the outlet}$$

Pressure fixed and velocity gradients zero — the outflow leaves without reflecting. This is also where you fix the pressure gauge.

$$u(y) = \frac{3}{2} U_{avg} \left[1 - \left(\frac{2y}{H}\right)^2\right]$$

Standard CFD practice

The Composite Loss for Flow

FIVE TERMS, AT MINIMUM

$$\mathcal{L} = \mathcal{L}_u + \mathcal{L}_v + w_c\,\mathcal{L}_{cont} + \mathcal{L}_{BC} + \mathcal{L}_{IC}$$

- two momentum residuals, one continuity residual, conditions on every surface
- **the continuity term is the one to watch** — different units from the momentum terms
- a stream function removes it entirely
- which is the strongest practical argument for that formulation
- and the no-slip condition deserves its own weight

TERM BY TERM

$\mathcal{L}_u, \mathcal{L}_v$	momentum	*always*
$\mathcal{L}^{cont}$	continuity	*removable by ψ*
$\mathcal{L}^{bc}$	no-slip, inlet, outlet	*weight the wall*
$\mathcal{L}_i^{c}$	initial field	*unsteady only*
$\mathcal{L}^{data}$	measurements	*inverse only*

Raissi et al. (2019); Liu Ch. 2.2.3 for weighting

PRACTICE

Non-Dimensionalisation Is Not Optional Here

CHANGE OF VARIABLES

$$\mathbf{u}^* = \frac{\mathbf{u}}{U}, \quad x^* = \frac{x}{L}, \quad p^* = \frac{p}{\rho U^2}$$

- **air has a viscosity around 1.5e-5, water around 1e-6** — raw numbers span orders
- scaled, all of that becomes one over the Reynolds number
- one number replaces four, and the fields become order one
- **note the pressure scaling** — by density times velocity squared, not by anything viscous
- train scaled, present unscaled, and state which you are showing

WATER IN A SMALL PIPE

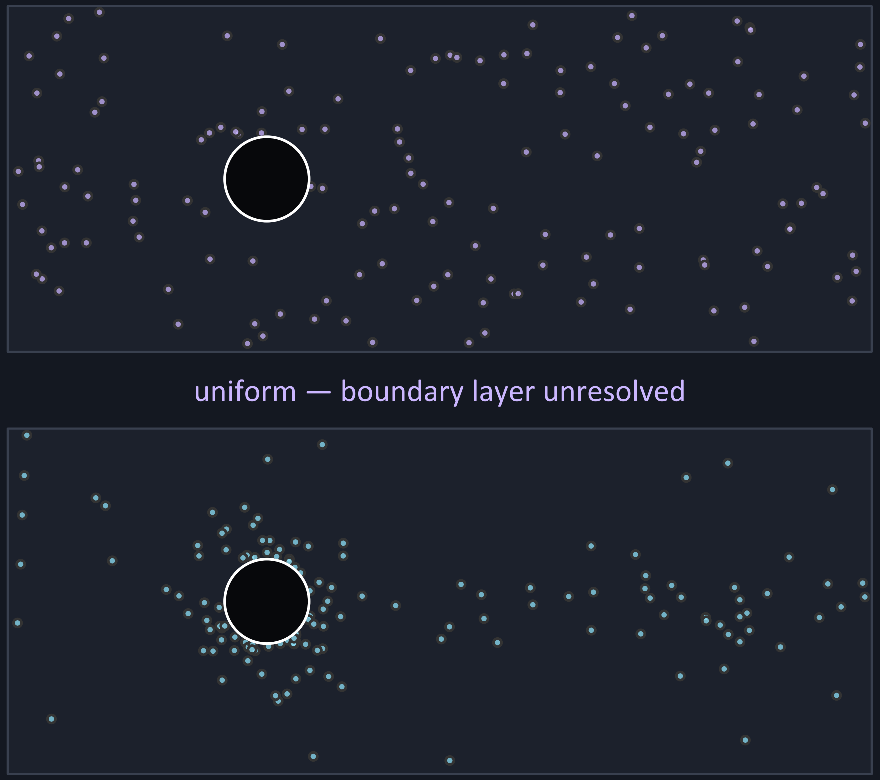
QUANTITY	PHYSICAL	SCALED
diameter D	0.05 m	**1**
velocity U	0.2 m/s	**1**
viscosity ν	1.0e−6	**1/Re**
Reynolds	—	**10 000**
pressure	ρU^2 = 40 Pa	**1**

Re = 10 000 in a real pipe — already turbulent, and already beyond a plain PINN. That gap is the subject of L9.2.

Where the Points Must Go

- flow is not uniformly difficult
- almost everything happens in the boundary layer and in the wake
- the boundary layer thins as the square root of Re
- so cluster points near walls and along the wake centreline
- residual-adaptive refinement works well here and is worth implementing

UNIFORM VS. GRADED SAMPLING



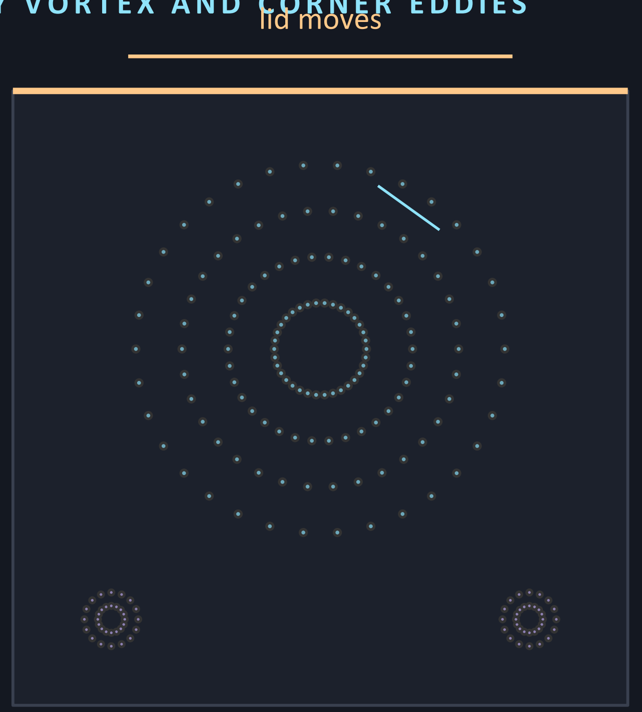
uniform — boundary layer unresolved

graded — points where the physics is

The Lid-Driven Cavity

- a square box, three fixed walls, and a lid sliding at constant speed
- **no inlet, no outlet, no obstacle** — the geometry could not be simpler
- and tabulated reference velocities exist across the centrelines
- it has one deliberate nastiness: the lid corners are singular
- **start here** — if the cavity fails, nothing downstream will work

PRIMARY VORTEX AND CORNER EDDIES



counter-rotating corner eddies — a real test of resolution

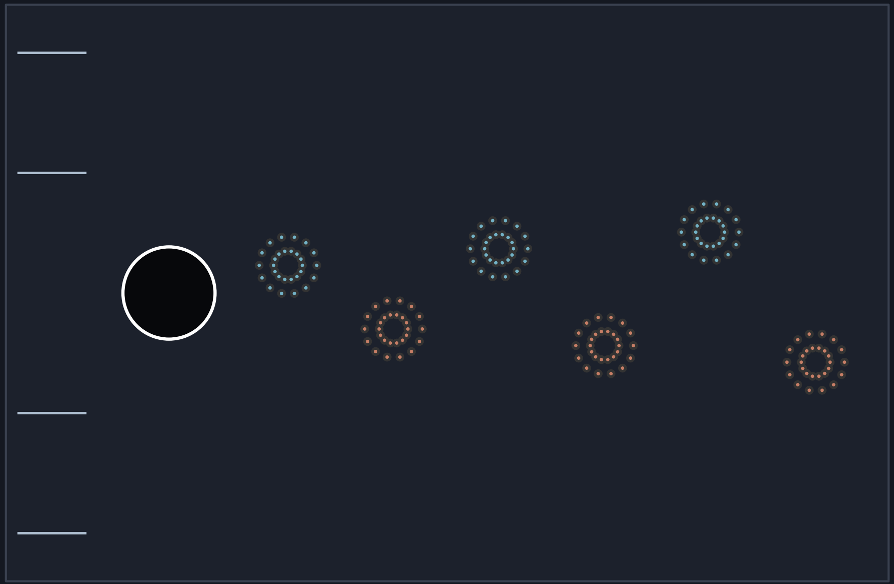
Flow Past a Cylinder at Re = 100

SHEDDING FREQUENCY

$$St = \frac{fD}{U} \approx 0.2 \quad (Re \sim 100)$$

- the canonical unsteady benchmark, and the one Raissi uses
- at Re = 100 the wake sheds periodically
- **unsteady but not chaotic** — one dominant frequency
- the Strouhal number is the verification target, not the field
- a model that reproduces the field but not the frequency has not solved the problem

THE KÁRMÁN VORTEX STREET



alternating vortices, shed at a single frequency

Raissi et al. (2019), NavierStokes.py

VERIFICATION

How to Know a Flow Solution Is Right

KOVASZNAY FLOW — AN EXACT STEADY SOLUTION OF NAVIER–STOKES

$$u = 1 - e^{\lambda x}\cos(2\pi y), \qquad \lambda = \frac{Re}{2} - \sqrt{\frac{Re^2}{4} + 4\pi^2}$$

EXACT SOLUTIONS

Kovasznay flow above, and Taylor–Green for the unsteady case. Both satisfy the full nonlinear equations, so they test everything at once.

REFERENCE TABLES

Ghia's cavity centreline velocities at Re = 100, 400, 1000. Compare pointwise; disagreement beyond a few percent is a bug.

INTEGRAL QUANTITIES

Drag and lift coefficients, and the Strouhal number for shedding. These are what an engineer reports, and they are sensitive to the boundary layer.

DIVERGENCE CHECK

Evaluate $\nabla \cdot u$ on a dense grid. With a stream function it must be at machine precision; with a penalty it tells you how well the constraint was enforced.

A pretty streamline plot proves nothing. Every one of these four is a number you can be wrong about.

Recovering Pressure From Velocity Alone

VISCOSITY AS A TRAINABLE PARAMETER

$$\nu \text{ trainable}, \qquad \mathcal{L} = \mathcal{L}_{PDE} + \mathcal{L}_{data}$$

- this is the result that made PINNs interesting to the fluids community
- **velocity can be measured** — by PIV, by ultrasound, by dye
- pressure inside a flow generally cannot
- train on scattered velocity with the residual in the loss, and pressure follows
- the same run recovers the viscosity to about one per cent from noisy data

WHAT GOES IN, WHAT COMES OUT

MEASURED
u, v at scattered points

IMPOSED
the N–S residual

RECOVERED
p, everywhere

RECOVERED
ν, to ~1%

pressure is never measured, and never needs to be

Raissi et al. (2019) — hidden fluid mechanics

Failure Modes in Laminar Flow

Continuity ignored	The divergence residual is orders of magnitude smaller than the momentum ones and gets outvoted.	Symptom: mass not conserved, streamlines that start or stop. Fix: stream function, or weight it.
Boundary layer unresolved	Too few points within $\delta \sim Re^{-1/2}$ of the wall for no-slip to bite.	Symptom: drag far too low, wake too weak. Fix: grade the sampling towards walls.
Trivial uniform flow	A constant velocity field satisfies the momentum equations and continuity exactly.	Symptom: the obstacle appears to have no effect. Fix: check the no-slip loss separately.
Pressure drift	Gauge freedom left unfixed, so absolute pressure differs run to run.	Symptom: irreproducible pressures. Fix: pin p at the outlet.
Re too high	Convection dominates, gradients steepen, and the smooth ansatz cannot follow.	Symptom: training stalls at a smooth, wrong field. This is the honest limit — see L9.2.

The third is the one students miss: a uniform field is a perfectly valid solution of the equations — just not of the boundary conditions.

Krishnapriyan et al. (2021); practice

PERSPECTIVE

PINN or CFD for a Laminar Flow?

USE CFD

A single forward solve on fixed geometry with known properties.
Anything above Re ~ 10^3.
You need a validated answer with grid-convergence evidence.
Forty years of verification behind it.

USE A PINN

Reconstructing a field from sparse measurements — the PIV problem.
Recovering pressure or viscosity that cannot be measured.
Geometry or parameters as inputs, so one training serves a sweep.

- for a laminar forward solve, a finite-volume code wins
- faster, more accurate, and far better verified

Judge the method by what it makes possible, not by whether it beats a mature solver at the solver's own job.

OVER TO YOU

Questions

TEN TO TEST WHETHER TODAY LANDED

- which term causes the difficulty, and why?
- what does continuity constrain, given it has no time derivative?
- what does the Reynolds number tell you about your chances?
- what does a stream function guarantee, and what does it cost?
- why is the pressure only determined up to a constant?
- where do the collocation points belong, and why not uniformly?

AND FOUR MORE

seven	why does the cavity come before any obstacle problem?
eight	what verifies a cylinder result, if not the field?
nine	how is pressure recovered when it was never measured?
ten	when would you use a finite-volume code instead?

THE ONE WITH NO SETTLED ANSWER

The lid corners are singular. What should the network do there?

BEFORE L9.2

Ex_09.1 notebooks 01 to 03. Bring the cavity centreline profile.

EXERCISE

Ex_9.1 — 2-D Burgers with an Interactive Interface

TASKS

1. Solve 2-D Burgers for the coupled field (u, v) on the unit square with a known exact solution.
2. Use the control panel to vary the collocation count and record how the error responds.
3. Sweep the Reynolds number upward and find where your model stops converging.
4. Compare a shared-trunk network against two separate networks at equal parameter count.
5. Generate the automatic report and interpret every figure in it.

QUESTIONS — AND WHERE THIS DECK ANSWERS THEM

Why is convection the source of every difficulty?	sl. 3, 6
Why does Re measure difficulty as well as physics?	sl. 5
Why does a stream function remove a loss term?	sl. 8, 9
Why is pressure only determined up to a constant?	sl. 10
Why does uniform sampling under-resolve a wall?	sl. 14

The interface lets you change parameters without touching code — but every question above needs an explanation, not a screenshot.

LOOKING AHEAD

What L9.2 Has to Deal With

- everything here assumed the flow is smooth, and below a few hundred it is
- above the transition it becomes chaotic and multiscale
- a PINN on the raw equations does not converge to the turbulent solution
- it converges to something smooth, and wrong
- **so L9.2 changes the question** — model turbulence rather than resolve it

SCALE RANGE GROWS WITH RE

The ratio of largest to smallest eddy grows as Re^(3/4). At Re = 10^4 that is a factor of a thousand in each direction — which is why turbulence is expensive for every method, not just this one.

SUMMARY

Key Takeaways

1	**Convection is the difficulty**	Delete it and flow behaves like heat. Keep it and you have nonlinearity, coupling and a Reynolds number that measures how hard the problem is.
2	**Continuity is a constraint**	It has no time derivative and no source. The pressure is the multiplier that enforces it — which is why pressure has no absolute value.
3	**Hard enforcement again**	A stream function makes $\nabla \cdot u = 0$ exact and deletes a loss term. The same argument as Dirichlet in L7, initial conditions in L8, orthogonality in L7.2.
4	**Sample where the physics is**	Boundary layers thin as Re grows. Uniform sampling silently fails to resolve them, and drag is the first thing to go wrong.
5	**The win is assimilation**	For a forward laminar solve, use CFD. For pressure from velocity, or viscosity from noisy data, there is no classical equivalent.

Next — L9.2: Turbulent Flow, where the question changes from solving to modelling.